Lecture 27 — Inverses and solving

Mon Nov 2

Last updated: October 19, 2026.

Boyd video (optional): Lecture 30 — left & right inverses · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 11.1–11.3. Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, you should be able to:

1. Interpret an inverse as undoing a transformation.
2. Determine whether a simple matrix is invertible.
3. Connect invertibility to uniqueness of $Ax = b$.
4. Explain why solving $Ax = b$ is generally preferable computationally to explicitly calculating A^{-1}.

Warm-up

Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}, \qquad x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

(a) Compute Ax.

(b) Can you find a matrix B such that $B(Ax) = x$? (Guess and check is fine for 2×2.) What would such a B "undo"?

Fill in

A square matrix A is **invertible** (nonsingular) if there exists a matrix A^{-1} of the same size with

$$A^{-1}A = AA^{-1} = _____.$$

Then A^{-1} is the **inverse** of A. Think of $x \mapsto Ax$ as a map: A^{-1} is the map that _____ it.

Solving. If A is invertible, the unique solution of $Ax = b$ is

$$x = _____.$$

Equivalently: $Ax = b$ has a unique solution for every b if and only if A is invertible.

Quick 2×2 test. For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$,

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

provided $ad - bc \neq 0$. (We will call $ad - bc$ the determinant on Friday.)

Compute carefully. Forming A^{-1} explicitly and then multiplying $A^{-1}b$ is usually _____ than solving $Ax = b$ by elimination / a solver. Prefer `A\b` / `solve(A,b)`, not "invert then multiply," unless you truly need many right-hand sides and already have a factorization.

Worked in class

W1. Undo a map. Let $A = \mathrm{diag}(2, 1/2)$. What does $x \mapsto Ax$ do geometrically? Write A^{-1} and check $A^{-1}A = I$.

W2. Invertible or not? For each matrix, decide and give a one-line reason (try to solve $Ax = 0$, or use the 2×2 formula).

(a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

W3. Unique solution. Suppose A is invertible and $Ax = b$ and $Ay = b$. Why must $x = y$?

W4. Solve, don't invert. For

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 4 \end{bmatrix},$$

solve $Ax = b$ by back substitution (no inverse needed). Then, if time, form A^{-1} and check $A^{-1}b$ matches.

W5. Cost intuition. You need x for one b. Which is the better default plan: (i) compute A^{-1} then $A^{-1}b$, or (ii) solve $Ax = b$ with a solver? One sentence.

Your turn

Y1. Let $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (90° CCW). What matrix undoes R? Check by multiplying.

Y2. Is $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ invertible? Explain using $Ax = b$ for a well-chosen b, or using $Ax = 0$.

Y3. Suppose A is invertible and $Ax = b$. True or false: if you change b to a different vector b', there is still exactly one solution x'.

Y4. True or false, with a one-line reason: for a square system, computing $x = A^{-1}b$ by first forming A^{-1} by hand is the recommended computational method in this course.

Answers (Your turn)

Y1. Rotation by 90° clockwise undoes CCW 90°, so

$$R^{-1} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

(equivalently R^{T}, or R^3). Check: $R^{-1}R = I$.

Y2. **Not invertible.** Columns are dependent (second column is zero), so $Ax = 0$ has nontrivial solutions, e.g. $x = (0, 1)$. Equivalently, $Ax = b$ fails for $b = (0, 1)$.

Y3. **True.** Invertibility means $Ax = b'$ has the unique solution $x' = A^{-1}b'$ for every right-hand side.

Y4. **False.** Prefer solving $Ax = b$ (elimination / a solver). Explicit A^{-1} is usually more work and less stable for a single b.